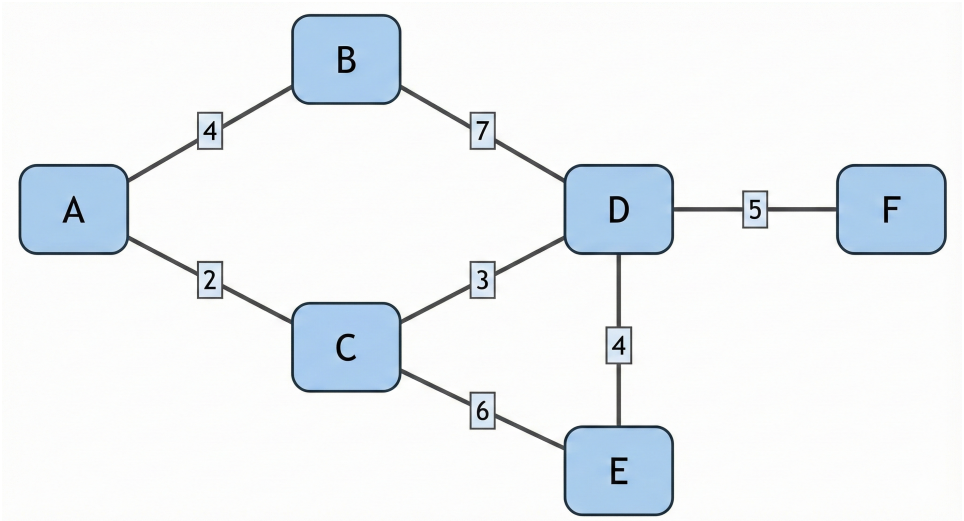
RUNI – Algorithms – Homework 6

Due date: Sunday ??/12/2025

NOTE: the general homework guidelines published in homework 1 are valid for all homework assignments in this course.

Note: Some of the questions are marked with ♠. It is recommended to work on them after the lecture and recitation of next week (we will cover some of the material only then).

Problem 1 (25 pts)



Answer the following questions (no need to justify your answers):

1. What is the weight of every MST in the graph? 18
2. How many different MSTs does the graph have? 1
3. Kruskal's algorithm is performed on the graph. What are the Union/Find sets after the three lightest edges are considered? break ties arbitrarily.

{A,C,D,E}, {B}, {F}

4. ♠ Assume Prim's algorithm is performed starting from vertex A. Show the best-edge array after two edges are added to the tree.

vertex	a	b	c	d	e	f
to		A			D	D
cost		4			4	5

5. Will your answers to (3) or (4) change if the weight of the edge CD would be 20 instead of 3?

(3) – {A,C,B}, {D,E}, {F}

(4) -

vertex	a	b	c	d	e	f
to				B	C	
cost				7	6	

Problem 2 (25 pts.)

Definition: Given an undirected connected graph G=(V,E), and its spanning tree, T, a vertex v is a HUB if its degree in T is identical to its degree in G.

Let G=(V,E) be a simple undirected connected graph with positive weights on the edges. Given are two vertices a,b∈V.

a. Suggest an algorithm for finding a minimum spanning tree $T_1=(V,E_1)$ in which a,b are HUBs (if no such MST exists, the algorithm should announce it). Justify shortly the correctness of your algorithm (no need to prove formally or to analyze time complexity).

Run Kruskal's algorithm with the following rule: in the sorting, if tie-breaking is required, give higher priority to edges adjacent to a and b.

Another formulation: Color yellow the edges adjacent to a and b, and find the most-yellow MST (see the recitation with this problem).

Let T be the MST output by Kruskal's algorithm. Check whether a and b are HUBs. If they are, return T, else announce that no MST in which a and b are HUBs exists.
Correctness: Similar to the most-yellow MST – the algorithm returns an MST with a maximal number of edges adjacent to a and b. Thus, if an MST in which a and b are HUBs exists, it will be found.

b. Suggest an algorithm for finding a spanning tree $T_2=(V,E_2)$ whose weight is minimal among all spanning trees in which a,b are HUBs (if no such ST exists, the algorithm should announce it). Justify shortly the correctness of your algorithm (no need to prove formally or to analyze time complexity). Note that T_2 is not necessarily an MST of G.

Set to 0 the weights of all edges adjacent to a and b.

Run Kruskal's algorithm on the resulting graph. If the edges adjacent to a and b form a cycle, return that no ST in which a and b are HUBs exists. Otherwise, complete the running and return the resulting MST.

Correctness: By reducing to 0 the weight of the edges adjacent to a and b, we give them highest priority. It is possible that no ST in which a and b are HUBs exists (if the edges adjacent to a and b form a cycle - for example when deg(a) +deg(b)>n). If this is not the case, then we complete the construction of the ST by adding edges of minimum total weight.

Problem 3 (25 pts)

Let $G = (V, E)$ be an undirected connected weighted graph.
For a spanning tree T of G, let $MaxCost(T)$ be the maximum weight of an edge in T.
An *economic tree* R of a graph G is a spanning tree of G with

$$MaxCost(R) = \min(\{MaxCost(T): T \text{ is a spanning tree of } G\}.$$

Prove or give a counter example for each of the following claims:

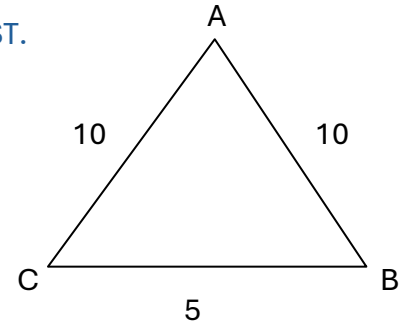
1) If T is an MST of G, then T is an *economic tree* of G.

Assume toward a contradiction that there is an MST T which is not an economic tree. Let R be an economic tree of G. Let e be a most expensive edge in T (which is more expensive than any other edge in R). Let A, B be the cut defined by removing e from T. Since R is connected (R is a tree), it must include an edge e' connecting A and B. If we replace e with e' in T we will get a connected graph, T', with $|V| - 1$ edges. Therefore, T' is a spanning tree of G. Since $w(e)' < w(e), w(T') < w(T)$, which is a contradiction to T being an MST.

2) If T is an *economic tree* of G, then G is an MST of G.

Counter example:

R={(a,c),(a,b)} is an economic tree of G but not an MST.



♠ **Problem 4** (25 pts.)

Let G=(V,E) be a connected undirected graph with positive integer edge costs, $c: E \to N^+$.
Assume that for two vertices u,v it holds that $(u, v) \notin E$.
Suggest an efficient algorithm that given a value x, returns whether the cost of an MST of G will reduce if an edge (u,v) whose cost is x is added to the graph.
The time complexity of your algorithm should be O(|E|).
Describe the algorithm, analyze and justify the time complexity, and explain its correctness shortly (no need to prove formally).

The algorithm:

1. Let G' = (V', E') be a new graph where V=V', ignoring vertices with no adjacent edges and $E' = \{ e \in E | w(e) \leq x \}$
2. Run DFS from u on G'
3. If u and v are connected return no, otherwise return yes

Time complexity:

O(|E|) to create G'

O(|E|) to run DFS

O(1) to return answer

Total: O(E|)

Explanation:

An MST does not have any cycles. If there is a path between u and v without any edges with cost more than x, then there is an MST that does not include (u,v). Otherwise, connecting the components of u and v requires at least one edge with cost more than x, and therefore adding the edge (u,v) with cost x results in a lower cost spanning tree.